



Modelling Financial Markets

Seminar 16: Digital Assets and DeFi — Hands-on



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Acronyms used in this material

- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ BNB: BNB (native token of BNB Chain, formerly Binance Coin)
- ▣ CAPM: Capital Asset Pricing Model
- ▣ CI: Confidence Interval
- ▣ CMKT: Crypto Market factor (value-weighted market excess return)
- ▣ CMOM: Crypto Momentum factor (past winners minus losers)
- ▣ CRIX: CRyptO Index
- ▣ CSMB: Crypto Size factor (small minus big coins)
- ▣ DAI: Dai (stablecoin of the Maker/Sky protocol)
- ▣ DTB3: 3-Month Treasury Bill: secondary market rate (FRED series code)
- ▣ ETF: Exchange-Traded Fund
- ▣ ETHA: iShares Ethereum Trust ETF (ticker)
- ▣ FRED: Federal Reserve Economic Data
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ KPSS: Kwiatkowski–Phillips–Schmidt–Shin test
- ▣ LVR: Loss-Versus-Rebalancing
- ▣ MSTR: Strategy Inc., formerly MicroStrategy (Nasdaq ticker)
- ▣ OLS: Ordinary Least Squares
- ▣ SE: Standard Error
- ▣ SSR: Sum of Squared Residuals
- ▣ TAR: Threshold AutoRegression
- ▣ US: United States
- ▣ USDC: USD Coin (stablecoin issued by Circle)
- ▣ XRP: XRP (the native token of the XRP Ledger)

Today's Question and Route

- Question of the seminar: **can the main numbers on digital assets be reproduced from the data, and how precise are they?**
- Route
 - ▶ First the primer “What You Need for Today” (four slides): [Chapter 16](#) comes after the seminar
 - ▶ A1 and A3 on paper: an index divisor, the no-arbitrage band of a pool
 - ▶ B1 and A7: Bitcoin's weekend effect, how fast USD Coin returns inside its band
 - ▶ B5 and B7: how well IBIT tracks Bitcoin, when Strategy's Bitcoin beta changed
 - ▶ C2: find the errors in an AI answer; B8 at home; B3 (threshold AR and its bootstrap test) after the lecture
- What you produce: the A answers, one table and one figure per B exercise, the corrected C2 code, one interpretation sentence per exercise
- Notebook for today: [open the seminar notebook in Colab](#); each task names its notebook section

Exercise Map

Exercise	Question	Type	Model
A1	how does a divisor keep an index continuous?	Solved	–
A3	when does arbitrage against a pool pay, and how much?	Solved	–
B1	is Bitcoin's weekend lull new since the ETFs?	Solved	–
A7	how many days does USD Coin need to return inside its band?	Solved	–
B5	does IBIT track Bitcoin one for one, and is the fee visible?	Solved	–
B7	when did Strategy's Bitcoin beta change?	Solved	–
C2	what is wrong in an AI answer?	Proposed	B1
B8	when did Coinbase's Bitcoin beta change? (homework)	Proposed	B7
B3	is the USD Coin peg a threshold process? (after the lecture)	Solved	A7
Extension packs	A2, A4–A6, A8, B2, B4, B6, C1, C3 (optional, after the lecture)	Solved and Proposed (marked on each task)	named on each task

- ▣ **[Solved]**: full solution in the slides and in the notebook, a model to copy; **[Proposed]**: you solve it, the solution is discussed in class
- ▣ Part A: derivations on paper; Part B: estimation and inference on real data; Part C: open questions and AI critique

Data Used

Series	Price used	Calendar	Sample	Used in
Bitcoin, Ether, Solana	daily close	every day, 7 a week	2018 – 18 Sep 2026	B1, B2, B5
USD Coin, Tether, Dai	daily close and daily low	every day	2021 – 18 Sep 2026	B3, B4, A7
Spot ETFs (IBIT, ETHA)	adjusted close	US trading days	from launch (11 Jan 2024; 23 Jul 2024)	B5, B6
Strategy, Coinbase	adjusted close	Friday closes	16 Apr 2021 – 18 Sep 2026	B7, B8
S&P 500; Nasdaq 100	index: daily close; ETFs:	US trading days	as the paired asset	B1, B7, C1
ETF (QQQ), gold ETF (GLD), long US Treasury ETF (TLT)	adjusted close			
3-month Treasury bill	FRED DTB3	US business days	2021–2025	C2

- Daily prices of the course data, one series per asset (field used: close / adjusted close)
- Two series with different calendars (crypto 7 days, shares 5 days): align the *prices* on common trading days first, then compute returns

Setup: Run and Verify

```
import numpy as np, pandas as pd
URL = ('https://raw.githubusercontent.com/'
      'danpele/MFM/main/data/market/')
def load(sym, field='close'):
    return pd.read_csv(URL + sym + '.csv',
                       index_col='date', parse_dates=True)[field]
btc = load('BTC-USD.CC').loc['2018-01-01':'2026-09-18']
r = 100 * np.log(btc).diff().dropna()
print(len(r), r.index[0].date(), r.index[-1].date())
```

- Expected output: 3 182 2018-01-02 2026-09-18
- Crypto trades 7 days a week: 3 182 daily returns in 8.7 years, about 365 a year; annualise with $\sqrt{365}$, not $\sqrt{252}$
- Further checks: USD Coin 2 087 daily closes from 1 Jan 2021; IBIT 673 common trading days and 140 weeks; Strategy 284 weeks
- If a number of observations differs, stop and check the asset, the price used and the dates

Notation Used in Every Exercise

- ▣ P_t : price on day t ; log return in %: $r_t = 100 \ln(P_t/P_{t-1})$; annualised volatility $\sigma_a = s\sqrt{m}$, $m = 365$ for crypto, 252 for shares, 52 for weekly data
- ▣ Peg deviation in basis points (1 bp = 0.01%): $d_t = 10^4(P_t - 1)$; half-life of an AR(1) with slope ϕ : $\ln 0.5 / \ln \phi$ days
- ▣ Constant-product pool with reserves x (Ether) and y (USD Coin): $xy = k$, pool price $p = y/x$, fee f , $\gamma = 1 - f$
Adams, Zinsmeister & Robinson (2020)
- ▣ Block bootstrap Künsch (1989): resample blocks of consecutive days (20–21), which keeps volatility clusters; HAC: Newey–West standard errors Newey & West (1987)
- ▣ sup-Wald Andrews (1993): the largest Wald statistic for a break over all candidate dates; its critical values are not χ^2

What You Need for Today (1/4): Crypto Indices and Liquidity Pools

- Chapter 16 comes after this seminar: these four slides are the minimum for today's core tasks; depth and evidence in the lecture
- Market-value index (CRIX, [Trimborn & Härdle \(2018\)](#)): $I_t = \sum_i P_{i,t} Q_{i,\tau} / D_\tau$, supplies Q fixed between reallocations; the divisor D sets the level
 - ▶ At a reallocation the basket changes, prices do not: choose the new D so that I is the same just before and just after
 - ▶ Example: one coin, $P = 100$, $Q = 10$, $D = 1$: $I = 1000$; Q becomes 12 at the same price: $D = 1200/1000 = 1.2$ keeps $I = 1000$
- Automated market maker (Uniswap v2, [Adams, Zinsmeister & Robinson \(2020\)](#)): a smart contract holds x Ether and y USD Coin and trades with anyone so that $xy = k$ stays constant; pool price $p = y/x$
 - ▶ Fee f on the input, $\gamma = 1 - f$: paying Δy buys Δx with $(x - \Delta x)(y + \gamma \Delta y) = k$; the fee stays in the pool
 - ▶ Example: $x = 10$, $y = 30\,000$ ($p = 3\,000$), $f = 0.3\%$; buying 1 Ether: $\gamma \Delta y = 300\,000/9 - 30\,000 = 3\,333.33$, $\Delta y = 3\,343.36$: the average price is above p (price impact plus fee)
 - ▶ Arbitrageurs trade against the pool when the outside price P moves away from p ; the fee creates a band where no trade pays (A3)

What You Need for Today (2/4): Stablecoins and a Threshold AR

- Stablecoin: a token meant to trade at 1 USD (USD Coin, Tether, Dai), redeemable or backed by reserves; arbitrage between the market and redemption pulls it back to par
 - ▶ The peg can break: USD Coin fell below 0.90 USD intraday in March 2023, when part of its reserves sat in a failed US bank; check the daily low, not only the close (C2)
- Deviation in bp: $d_t = 10^4(P_t - 1)$; arbitrage costs money, so small deviations may persist and large ones revert fast
- Threshold AR (TAR, [Balke & Fomby \(1997\)](#)): $d_t = \phi_{\text{in}}d_{t-1} + \varepsilon_t$ if $|d_{t-1}| \leq c$ (inside the band), $d_t = \phi_{\text{out}}d_{t-1} + \varepsilon_t$ otherwise
 - ▶ Without shocks, $d_n = \phi^n d_0$ in one regime; half-life $h_{1/2} = \ln 0.5 / \ln \phi$
 - ▶ Example: $\phi_{\text{out}} = 0.5$, $d_0 = -100$ bp, $c = 20$ bp: $100 \cdot 0.5^n \leq 20$ needs $n \geq \ln 0.2 / \ln 0.5 = 2.32$, so inside the band on day 3; half-life 1 day
 - ▶ Estimating c and testing $\phi_{\text{in}} = \phi_{\text{out}}$ needs the sup-Wald logic of slide 4/4 plus a bootstrap: B3, after the lecture

What You Need for Today (3/4): Spot ETFs and Tracking

- Spot Bitcoin ETF (IBIT, since 11 January 2024): a US-listed fund that holds Bitcoin; it trades on US exchange days, Bitcoin trades every day
- Tracking: $\ell_t = \ln(P_t^{\text{ETF}} / P_t^{\text{BTC}})$; a sponsor fee of 0.25% a year paid in Bitcoin makes ℓ_t drift down by about 0.25% a year BlackRock (2026)
 - ▶ Fee + timing noise: ℓ_t is trend-stationary; ADF Dickey & Fuller (1979) has H_0 unit root, KPSS Kwiatkowski et al. (1992) has H_0 (trend) stationarity: use both, they answer opposite questions
- Asynchronous closes: the ETF closes at 16:00 New York time, the daily Bitcoin price at a different hour; part of a Bitcoin move reaches the ETF the next day
 - ▶ The daily OLS beta is then below 1; the Dimson beta Dimson (1979) sums lag, contemporaneous and lead: $b_{-1} + b_0 + b_1$; weekly returns have the same effect
 - ▶ Example: a 10% Bitcoin move split 6% today and 4% tomorrow in the ETF: $b_0 \approx 0.6$, lag $b_{-1} \approx 0.4$, sum 1

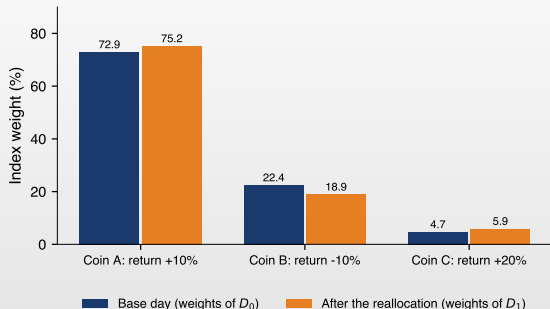
What You Need for Today (4/4): A Break at an Unknown Date

- Regression with a break at week τ : $y_t = x_t^\top \beta_1 + u_t$ for $t \leq \tau$, $x_t^\top \beta_2 + u_t$ after; $W(\tau)$: HAC Wald statistic for $H_0: \beta_1 = \beta_2$
- Known date (e.g. the ETF launch): compare $W(\tau)$ with χ_q^2 , q = number of coefficients that may change
 - ▶ But a date chosen after looking at the data is not known in advance, and an imposed split finds a “change” wherever the true break lies
- Unknown date: $\sup W = \max_\tau W(\tau)$ over the middle 70% of the sample (15% trimmed at each end) [Andrews \(1993\)](#)
 - ▶ The maximum of many Wald statistics is larger than any single one: its critical values are not χ^2 ; for B7 ($q = 3$) the 5% value is 13.95, against 7.81 for χ_3^2
 - ▶ Break date: the τ with the smallest total sum of squared residuals; its 95% CI [Bai \(1997\)](#)
- More in [Chapter 16](#), after the seminar: stylised facts of crypto returns, stablecoin runs, loss versus rebalancing in pools and the crypto factor case study

A1 [Solved]: A Three-Coin Index and Its Divisor — Task

- ▣ **Question:** how does a market-value index stay continuous when its quantities change?
- ▣ **Data and inputs**
 - ▶ Coins A, B, C on the base day: prices $P_0 = (60\,000; 3\,000; 150)$ USD, circulating supplies $Q = (19.5; 120; 500)$ million
 - ▶ Laspeyres index $I_t = \sum_i P_{i,t} Q_{i,\tau} / D_\tau$, quantities fixed within the month (primer slide 1/4)
- ▣ **Tasks**
 1. Compute the market values, the weights and the divisor D_0 that sets $I_0 = 1000$.
 2. At month end the prices are $P_1 = (66\,000; 2\,700; 180)$: compute I_1 and check that $I_1/1000 - 1$ is the weighted average of the three returns.
 3. At the reallocation the supplies become $(19.52; 120.1; 560)$ million: compute the new divisor D_1 that keeps I_1 unchanged, and the new weights.
- ▣ **Report:** D_0 , I_1 , D_1 and both sets of weights, with every intermediate product
- ▣ **Interpretation:** why must the divisor change at a reallocation, and what would an investor see without the change?
- ▣ Notebook: section A1

A1: Solution



- 1. $M_i = P_{i,0} Q_i$: 1 170, 360, 75 billion USD, total 1 605; weights $w_i = M_i / \sum_j M_j$: 72.9%, 22.4%, 4.7%;
 $D_0 = \sum_j M_j / 1000 = 1.605 \cdot 10^9$
- 2. $I_1 = \sum_i P_{i,1} Q_i / D_0 = 1059.81$; returns +10%, -10%, +20%; check:
 $72.9\% \cdot 10\% - 22.4\% \cdot 10\% + 4.7\% \cdot 20\% = 5.98\%$
- 3. $D_1 = \sum_i P_{i,1} Q_i^{\text{new}} / I_1 = 1.6167 \cdot 10^9$; new weights 75.2%, 18.9%, 5.9% (chart); the level stays 1059.81
- Interpretation:** the divisor absorbs basket changes that are not price moves; without it the index would jump by the value of the added coins, a fake return

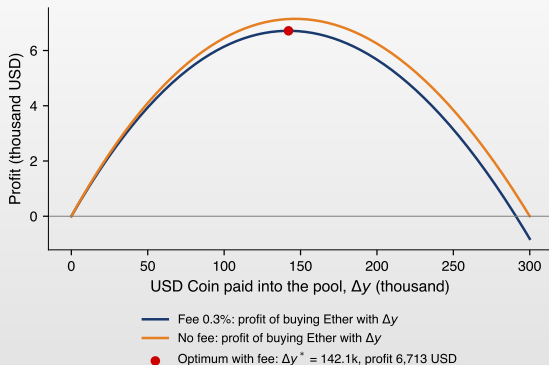
A3 [Solved]: The No-Arbitrage Band of a Pool with Fees — Task

- **Question:** when does it pay to trade against a pool, and how large is the optimal trade?
- **Data and inputs**
 - ▶ Constant-product pool with $x = 1\,000$ Ether and $y = 3\,000\,000$ USD Coin: $k = 3 \cdot 10^9$, pool price $p = 3\,000$
 - ▶ Fee $f = 0.3\%$ charged on the input and kept in the pool (Uniswap v2); outside price P ; one trade, no transaction cost on the blockchain
- **Tasks**
 1. Derive the marginal cost of buying and the marginal revenue of selling Ether in the pool, and show that no trade pays while $\gamma P \leq p \leq P/\gamma$.
 2. For $P = 3\,300$, write the profit $\pi(\Delta y)$ of buying Ether with Δy USD Coin, solve $\pi'(\Delta y) = 0$ and compute Δy^* , the Ether bought and the new pool price.
 3. Compute the arbitrage profit with and without the fee.
- **Report:** the band, Δy^* , Δx , the new pool price and both profits
- **Interpretation:** where does arbitrage leave the pool price, and who pays the profit?
- **Notebook:** section A3

A3: Solution (1/2)

- ▣ 1. Buying Δx Ether: $(x - \Delta x)(y + \gamma \Delta y) = k$ gives $\Delta y = \frac{1}{\gamma} \left(\frac{k}{x - \Delta x} - y \right)$; at $\Delta x = 0$ the marginal cost is $k/(\gamma x^2) = p/\gamma$
- ▣ Selling Ether: $(x + \gamma \Delta x)(y - \Delta y) = k$ gives the marginal revenue γp
- ▣ Buying pays iff $p/\gamma < P$, selling iff $\gamma p > P$; neither while $\gamma P \leq p \leq P/\gamma$, here $[3290.10, 3309.93]$ for $P = 3\,300$: the pool price 3000 is below the band, so buying Ether pays

A3: Solution (2/2)

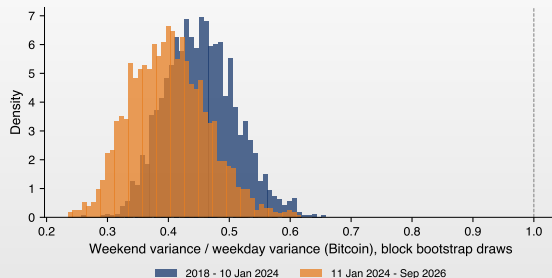


- 2. $\pi(\Delta y) = P[x - k/(y + \gamma\Delta y)] - \Delta y$; $\pi' = 0$ gives the effective reserve
 $y^e = y + \gamma\Delta y^* = \sqrt{\gamma Pk} = 3\,141\,703$
- $\Delta y^* = (y^e - y)/\gamma = 142\,130$ USD Coin buys
 $\Delta x = 45.10$ Ether ($x' = k/y^e = 954.90$); the fee stays in the pool: $y' = y + \Delta y^*$, new price $y'/x' = 3290.55$
- 3. Profit $P\Delta x - \Delta y^* = 6\,713$ USD; without the fee 7 147 USD (chart)
- **Interpretation:** arbitrage stops at the edge of the new band ($P \in [3280.67, 3300.45]$), not at P ; the profit is paid by the liquidity providers, who sold Ether below the market price

B1 [Solved]: Bitcoin's Volatility and the Weekend Effect — Task

- ▣ **Question:** is Bitcoin's weekend lull (lower variance on Saturdays and Sundays) a new effect of the spot ETFs?
- ▣ **Data and inputs**
 - ▶ Bitcoin daily log returns, 2018-01-02 – 18 Sep 2026, 3 182 days, 7 a week; S&P 500 on its own calendar
 - ▶ Split at the first spot ETF day, 11 January 2024: 2 200 days before, 982 after; moving-block bootstrap [Künsch \(1989\)](#), 2 000 resamples
- ▣ **Tasks**
 1. Compute the annualised volatility $s\sqrt{365}$ with a 95% block-bootstrap interval (blocks of 20 days) and its ratio to the S&P 500 volatility.
 2. Compute the ratio of weekend to weekday return variance before and after the ETFs, each with a block-bootstrap interval (blocks of 21 days).
 3. Compute a 95% interval for the change in the ratio from the difference of the two bootstrap distributions.
- ▣ **Report:** the volatility with its interval, both ratios with intervals, the change with its interval, the bootstrap histogram
- ▣ **Interpretation:** is the weekend lull new since the ETFs, and why resample blocks rather than single days?
- ▣ Notebook: section B1

B1: Solution

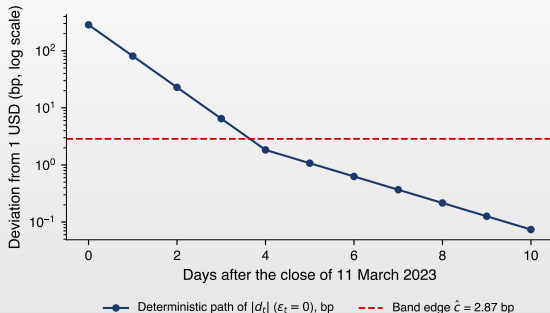


- 1. Resample 2000 series of 20-day blocks, recompute $s\sqrt{365}$, take the 2.5% and 97.5% percentiles: 64.1%, CI [58.2, 70.4]; S&P 500 19.2%, ratio 3.3
- 2. Before: 0.45 [0.35, 0.57]; after: 0.39 [0.29, 0.53] (chart: the two bootstrap distributions)
- 3. Change -0.05 , CI $[-0.22, 0.11]$: contains 0, so no significant change at 5%
- Interpretation:** weekends carry less than half the weekday variance before and after 2024: the lull is not ETF-driven; blocks keep the volatility clusters that single-day resampling destroys
- Caveat: with $\hat{\alpha}_H$ near 2.7 (Chapter 16, after the seminar), $Er^4 < \infty$ is doubtful and a variance bootstrap need not be consistent Athreya (1987): the intervals are indicative

A7 [Solved]: Threshold AR Algebra — Task

- ▣ **Question:** how many days does USD Coin need to return inside its band after the March 2023 shock?
- ▣ **Data and inputs**
 - ▶ The threshold AR (primer slide 2/4) with its USD Coin estimates from B3 (after the lecture): $c = 2.8650$ bp, $\phi_{\text{in}} = 0.5855$, $\phi_{\text{out}} = 0.2832$
 - ▶ Deterministic path: all shocks set to zero, $\varepsilon_t = 0$; start at $d_0 = -285$ bp, the close of 11 March 2023
- ▣ **Tasks**
 1. Derive the half-life in each regime and compute it.
 2. Derive and compute the number of days until $|d_n| \leq c$.
 3. State what $\phi_{\text{in}} = 1$ would mean for the dynamics inside the band.
- ▣ **Report:** two half-lives in days, the number of days of task 2, one sentence on task 3
- ▣ **Interpretation:** what does $\phi_{\text{in}} < 1$ say about trading inside the band?
- ▣ **Notebook:** section A7

A7: Solution



- 1. In one regime $d_h = \phi^h d_0$; $\phi^h = 0.5$ gives $h_{1/2} = \ln 0.5 / \ln \phi$: 1.29 days inside, 0.55 days outside
- 2. $|d_0| \phi_{\text{out}}^n \leq c \iff n \geq \ln(c/|d_0|) / \ln \phi_{\text{out}} = \ln(2.8650/285) / \ln 0.2832 = 3.65$: inside the band after 4 days, then the slower inside dynamics (chart)
- 3. $\phi_{\text{in}} = 1$: a random walk inside the band (the band-TAR of [Balke & Fomby \(1997\)](#)), no pull towards par until the deviation leaves it
- Interpretation: $\phi_{\text{in}} < 1$ means reversion even inside the band; market makers are one hypothesis, testable only with transaction-level data

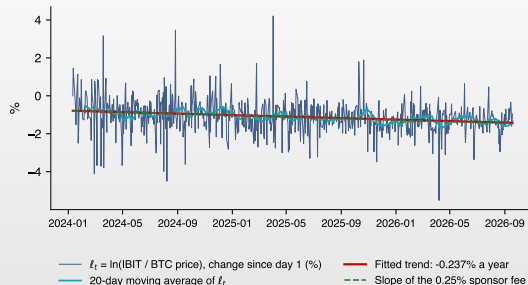
B5 [Solved]: IBIT: Tracking, Timing and the Fee Drift — Task

- **Question:** does IBIT track Bitcoin one for one, and is its 0.25% fee visible in the prices?
- **Data and inputs**
 - ▶ IBIT (adjusted close) and Bitcoin on IBIT's trading days, 11 Jan 2024 – 18 Sep 2026: 673 days, 140 Friday-to-Friday weeks; prices aligned on common trading days first, then returns
 - ▶ $\ell_t = \ln(P_t^{\text{IBIT}}/P_t^{\text{BTC}})$; sponsor fee 0.25% a year, paid in Bitcoin [BlackRock \(2026\)](#); level 5%
- **Tasks**
 1. Estimate the daily and the weekly beta with Newey–West SE, and the Dimson sum beta $b_{-1} + b_0 + b_1$ with one lag and one lead of Bitcoin [Dimson \(1979\)](#).
 2. Run an ADF test [Dickey & Fuller \(1979\)](#) and a KPSS test [Kwiatkowski et al. \(1992\)](#), both with trend, on ℓ_t .
 3. Estimate the annual drift two ways, each with a 95% CI: the mean of $\Delta\ell_t$ (HAC) and the slope of ℓ_t on time in years (HAC, 20 lags).
- **Report:** three betas with SE, both test statistics with decisions, both drifts with CIs, the chart of ℓ_t
- **Interpretation:** why is the daily OLS beta below 1, and which drift estimator is precise?
- Notebook: section B5

B5: Solution (1/2)

- ▣ 1. Daily OLS $\hat{\beta} = 0.93$; weekly $\hat{\beta} = 1.007$ (SE 0.011), $R^2 = 0.98$
- ▣ Dimson: $\hat{b}_{-1} = 0.05$, $\hat{b}_0 = 0.94$, $\hat{b}_1 = 0.01$; sum 1.00 (SE 0.023), $t(\beta = 1) = -0.07$: not different from 1
- ▣ Why below 1: Bitcoin's daily close and the New York close are hours apart, part of each move reaches IBIT a day later; the lag term restores $\beta = 1$, as weekly data do
- ▣ 2. ADF = -11.07 , $p < 0.001$: reject a unit root; KPSS = 0.041, $p > 0.10$: do not reject trend stationarity. So $\ell_t = \text{fee trend} + \text{stationary timing noise}$

B5: Solution (2/2)



- 3. Mean of $\Delta \ell_t$ per year: -0.36% , CI $[-9.28, 8.57]$: it telescopes to $(\ell_T - \ell_0)/T$ and uses two noisy days only
- Trend slope (red line): -0.237% a year, CI $[-0.315, -0.158]$: excludes 0, contains -0.25% (green line)
- Under a trend-stationary ℓ_t the trend slope with HAC SE is consistent and its CI valid; with a unit root the mean of differences would be the right tool (a pretest does not guarantee coverage)
- **Interpretation:** IBIT tracks Bitcoin one for one, and the 0.25% fee is visible as a slow downward trend in ℓ_t

B7 [Solved]: When Did Strategy's Bitcoin Beta Change? — Task

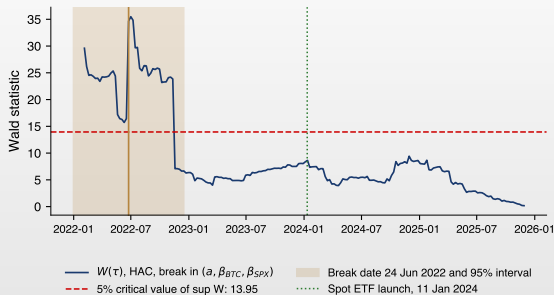
- **Question:** did Strategy's exposure to Bitcoin change, and did the data choose the ETF launch as the date?
- **Data and inputs**
 - ▶ Weekly (Friday) log returns of Strategy (MSTR), Bitcoin and the S&P 500, 16 Apr 2021 – 18 Sep 2026, 284 weeks; prices aligned on common trading days first, then returns
 - ▶ $r_t^{\text{MSTR}} = a + \beta_{\text{BTC}} r_t^{\text{BTC}} + \beta_{\text{SPX}} r_t^{\text{SPX}} + u_t$, Newey–West SE; level 5%
- **Tasks**
 1. Estimate the model on the full period and on the two subperiods split at the ETF launch (to 5 Jan 2024; from 12 Jan 2024), and test $\beta_{\text{BTC}} = 1$ in each.
 2. For every week in the middle 70% of the sample, compute the HAC Wald statistic $W(\tau)$ for a break in all three coefficients, and compare sup W with its 5% critical value [Andrews \(1993\)](#).
 3. Estimate the break date by least squares with its 95% CI [Bai \(1997\)](#), and the coefficients in each regime.
- **Report:** the table of task 1, sup W with the critical value and decision, the date with its CI, the chart of $W(\tau)$
- **Interpretation:** why is a split at an imposed date not a test of when the beta changed?
- **Notebook:** section B7

B7: Solution (1/2)

	n	$\hat{\beta}_{\text{BTC}}$ (SE)	$t(\beta_{\text{BTC}} = 1)$	$\hat{\beta}_{\text{SPX}}$ (SE)	alpha, % a year (t)
Apr 2021 – Sep 2026	284	1.18 (0.11)	1.64	1.02 (0.24)	−4.5 (−0.19)
Apr 2021 – 5 Jan 2024	143	1.01 (0.14)	0.08	1.40 (0.35)	−0.7 (−0.02)
12 Jan 2024 – Sep 2026	141	1.50 (0.12)	4.12	0.48 (0.29)	−9.3 (−0.29)

- ▣ 1. $\hat{\beta}_{\text{BTC}}$ rises from 1.01 to 1.50; $t = 4.12$ tests $\beta_{\text{BTC}} = 1$ after 2024, not a change
- ▣ The change itself: a pooled regression with post-2024 interactions gives $\hat{\beta}_{\text{post}} - \hat{\beta}_{\text{pre}} = 0.49$ (HAC SE 0.18, $t = 2.69$); alpha is never significant

B7: Solution (2/2)



- 2. $\sup W = 35.49$ against a 5% critical value of 13.95, $p < 0.001$: reject stability at 5%
- 3. Break 24 Jun 2022, 95% CI 31 Dec 2021 – 16 Dec 2022; W at the ETF launch only 8.64
- Regimes: Bitcoin beta $0.69 \rightarrow 1.48$, S&P 500 beta $1.25 \rightarrow 1.00$
- **Interpretation:** the data choose mid-2022, a period of crypto-wide stress, not the ETF launch; an imposed split finds a change wherever the true break lies before it, so it cannot date the change

C2 [Proposed]: Critique an AI Answer — the Answer

- **Prompt** sent to an AI assistant: “Compare Bitcoin with lending USD Coin (USDC) at 5% a year: compute Bitcoin’s annualised Sharpe ratio on daily data for 2021–2025, with the USDC yield as the risk-free rate”
- **Answer:** the code below and the conclusion

```
import numpy as np, pandas as pd
btc = pd.read_csv('data/market/BTC-USD.CC.csv', index_col='date',
                  parse_dates=True)['close']
r = np.log(btc.loc['2021':'2025']).diff().dropna()    # daily log returns
mu = r.mean() * 252                                # annual mean
sd = r.std() * np.sqrt(252)                          # annual volatility
rf = 0.05                                           # USDC lending: risk-free
sharpe = (mu - rf) / sd
print(f"mu={mu:.1%}, vol={sd:.1%}, Sharpe={sharpe:.2f}")
```

- **Conclusion of the AI:** “mu = 15.1%, vol = 48.5%, Sharpe = 0.21. USDC is fully backed by dollars and has never traded below \$0.99, so lending it is a risk-free investment”

C2 [Proposed]: Critique an AI Answer — Task

- ▣ **Model:** B1 (annualising crypto); the USD Coin daily close and low (primer slide 2/4)
- ▣ **Data:** Bitcoin (daily close), 2021–2025; USD Coin (daily close and low); 3-month Treasury bill, [FRED DTB3](#)
- ▣ **Tasks**
 1. Find the five substantive errors, three in the code and two in the text.
 2. For each, say how you would detect it: a count in the data, the data themselves, or a definition from the primer.
 3. Write the corrected code and report the annualised mean, volatility and Sharpe ratio.
 4. Log every error you found in the format of `AI_ERRORS.md`: request, answer, error, how found, correction.
- ▣ **Interpretation:** how many daily returns should 2021–2025 contain for Bitcoin, and if lending USDC were risk-free, why would it pay more than a Treasury bill?
- ▣ AI code usually runs without an error message: the mistakes are in the finance, not in the syntax

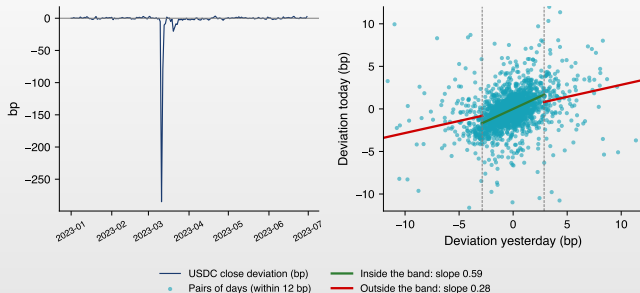
B8 [Proposed]: Coinbase — Task

- **Question:** is Coinbase a Bitcoin proxy or a US growth stock, and when did its Bitcoin beta change?
- **Model:** B7
- **Data and inputs**
 - ▶ Weekly log returns of Coinbase (listed April 2021), Bitcoin and the S&P 500, 16 Apr 2021 – 18 Sep 2026, 283 weeks
 - ▶ Same two-factor model, HAC settings, trimming and level as B7
- **Tasks**
 1. Estimate the model on the full period and the two subperiods, and test $\beta_{\text{BTC}} = 1$ in each.
 2. Compute $W(\tau)$ for every candidate week, plot it, and compare $\sup W$ with the 5% critical value.
 3. Estimate the break date with its 95% CI and the Bitcoin beta in each regime.
- **Report:** the table of task 1, the chart of $W(\tau)$, the test decision, the date with its CI
- **Interpretation:** do Strategy and Coinbase break in the same week, and what common event could explain it?
- **Notebook:** section B8

B3 [Solved]: Is the USD Coin Peg a Threshold Process? — Task

- **Question:** does USD Coin revert to 1 USD faster outside an arbitrage band than inside it (lecture, Section 4)?
- **Data and inputs**
 - ▶ USD Coin daily close, 1 Jan 2021 – 18 Sep 2026, 2 087 days; $d_t = 10^4(P_t - 1)$ in bp
 - ▶ Threshold AR without intercept [Balke & Fomby \(1997\)](#): $d_t = \phi_{\text{in}} d_{t-1} + \varepsilon_t$ if $|d_{t-1}| \leq c$, $d_t = \phi_{\text{out}} d_{t-1} + \varepsilon_t$ otherwise; level 5%
- **Tasks**
 1. Compute the standard deviation of d_t , the days with $|d_t| > 50$ bp, the lowest close and the lowest daily low.
 2. Estimate c by least squares over the values of $|d_{t-1}|$ that leave at least 15% of the days in each regime, and ϕ_{in} , ϕ_{out} with White standard errors.
 3. Test $H_0: \phi_{\text{in}} = \phi_{\text{out}}$ with $\sup_c W(c)$ and the fixed-regressor wild bootstrap of [Hansen \(1996\)](#) (999 draws); also report the χ^2_1 p -value.
 4. Repeat steps 2–3 without the pairs (d_{t-1}, d_t) that touch 9–16 March 2023.
- **Report:** $\hat{c}$, both slopes with SE and half-lives, $\sup W$, both p -values, with and without March 2023
- **Interpretation:** why is χ^2_1 wrong here, and does one stress week carry the evidence?
- Notebook: section B3

B3: Solution (1/2)



- Left: close deviation in early 2023; right: today against yesterday within ± 12 bp, slopes inside and outside $\pm \hat{c}$
- 1. s.d. 7.5 bp (3.7 without March 2023); 3 days beyond 50 bp; lowest close -285 bp, lowest low -1226 bp, both on 11 March 2023
- 2. For each c in $[0.3, 2.9]$ fit OLS in each regime, keep the smallest total SSR: $\hat{c} = 2.9$ bp (15.7% of days outside); $\hat{\phi}_{\text{in}} = 0.59$ (SE 0.04), $\hat{\phi}_{\text{out}} = 0.28$ (SE 0.03); half-lives 1.29 and 0.55 days

B3: Solution (2/2)

- 3. $W(c) = (\hat{\phi}_{\text{in}} - \hat{\phi}_{\text{out}})^2 / (\hat{v}_{\text{in}} + \hat{v}_{\text{out}})$ for each c ; $\sup W = 32.5$
- Bootstrap: $d_t^* = \hat{e}_t \eta_t$ ($\hat{e}_t$: residuals of the linear AR(1), $\eta_t \sim N(0, 1)$), regress on the same fixed d_{t-1} , recompute $\sup W^*$ 999 times: $p = 0.001$ (χ_1^2 : < 0.001): reject linearity at 5%
 - ▶ Multiplying each residual by η_t keeps its own variance: the sample excess kurtosis of d_t is 1031 (a descriptor of extreme days, not an estimate), calm and stress days must not be mixed
- 4. Without 9–16 March 2023: $\sup W = 5.5$, bootstrap $p = 0.291$: linearity no longer rejected; χ_1^2 would give 0.019, a false rejection
- Why χ_1^2 fails: c is not identified under H_0 , and the supremum over many c exceeds any single $W(c)$ (the Davies problem)
- **Interpretation:** the evidence for a band rests on one stress week; without it the data do not reject a linear, fast-reverting peg (non-rejection, not proof of linearity)

Extension Packs: What Each One Needs

Pack	Exercises	Prerequisites
1. Index divisor	A2 [Proposed]	A1
2. Weighted pools	A4 [Proposed], A5 [Solved], A6 [Proposed]	A3; lecture: LVR
3. Stablecoin pegs	A8, B4 [Proposed]	B3, A7
4. Crypto volatility	B2 [Proposed]	B1
5. Spot ETFs	B6 [Proposed]	B5
6. Hedge or risk asset?	C1 [Proposed]	B1, B5; lecture: Forbes–Rigobon
7. Crypto factors after 2020	C3 [Proposed], project seed	B5, B7; lecture case study

- Each exercise has its notebook section; solutions of [Proposed] exercises are discussed in class

A2 [Proposed]: Replacing a Constituent — Task

- ▣ **Question:** how must the divisor change when a coin leaves the index and another enters?
- ▣ **Model:** A1
- ▣ **Data and inputs**
 - ▶ The index of A1 after the reallocation: level $I = 1059.81$, divisor $1.6167 \cdot 10^9$, month-end prices of A1
 - ▶ On the same day coin C is delisted and coin D enters with $P_D = 0.5$ USD and $Q_D = 40$ billion coins
- ▣ **Tasks**
 1. Compute the market value of the old and of the new basket on the switch day.
 2. Derive the divisor that keeps I continuous and compute it.
 3. Compute the level the day after if the divisor were not changed and no price moved.
- ▣ **Report:** both basket values, the new divisor, the level of task 3
- ▣ **Interpretation:** why does the divisor fall here, and who would be misled by the fake loss?

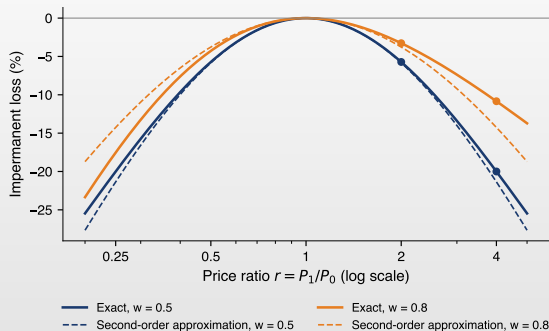
A4 [Proposed]: Arbitrage in a Weighted Pool — Task

- **Question:** does a pool that holds mostly Ether lose less to arbitrageurs than a 50/50 pool?
- **Model:** A3
- **Data and inputs**
 - ▶ Weighted pool (Balancer type) $x^w y^{1-w} = k$, $w = 0.8$, $x = 1\,000$ Ether, pool price $p = 3\,000$; fee 0.3% on the input, kept in the pool
 - ▶ The outside price rises to $P = 3\,300$; one arbitrage trade
- **Tasks**
 1. Differentiate the invariant to show that the pool price is $p = \frac{w}{1-w} \frac{y}{x}$, and compute y .
 2. Show that the no-arbitrage band $\gamma P \leq p \leq P/\gamma$ does not depend on w .
 3. Derive the reserves after the optimal trade, compute Δy^* , the profit, and the profit per unit of pre-trade pool value $px + y$; compare with A3 (6 million USD pool).
- **Report:** y , x' , Δy^* , the profit, the profit per unit of value and its ratio to A3
- **Interpretation:** why does a pool with $w = 0.8$ lose less to arbitrageurs?

A5 [Solved]: Impermanent Loss of a Weighted Pool — Task

- ▣ **Question:** how does the impermanent loss depend on the weight w of Ether in the pool?
- ▣ **Data and inputs**
 - ▶ A liquidity provider owns a pool $x^w y^{1-w} = k$ with pool price $p = \frac{w}{1-w} \frac{y}{x}$; arbitrage keeps p at the market price P , no fees
 - ▶ The price moves from P_0 to $P_1 = rP_0$; the benchmark is holding the initial tokens
- ▣ **Tasks**
 1. Show that the value shares stay w and $1 - w$ and that $V(P) = cP^w$.
 2. Derive $IL_w(r) = r^w / (wr + 1 - w) - 1$ and check that $w = 1/2$ gives $2\sqrt{r}/(1 + r) - 1$.
 3. Expand $\ln(1 + IL_w)$ to second order in $u = \ln r$ and compute IL_w for $r = 2, 4$ and $w = 0.5, 0.8$.
- ▣ **Report:** the formula, the approximation and four values
- ▣ **Interpretation:** which pool loses less when the price doubles, and why?

A5: Solution



- 1. $p = P$ gives $Px = wV$, $y = (1 - w)V$ with $V = Px + y$; in the invariant:
 $V = k w^{-w} (1 - w)^{w-1} P^w = cP^w$
- 2. Pool: $V_1/V_0 = r^w$; holding: $wr + 1 - w$; ratio minus 1 is IL_w ; at $w = 1/2$: $2\sqrt{r}/(1 + r) - 1$
- 3. $\ln(1 + IL_w) \approx -\frac{1}{2}w(1 - w)u^2$ (dashed); $r = 2$:
 -5.72% ($w = 0.5$), -3.27% ($w = 0.8$); $r = 4$:
 -20.00% , -10.84%
- Interpretation: the loss scales with $w(1 - w)$, the curvature of $V(P)$: a mostly-Ether pool rebalances less and sells less volatility

A6 [Proposed]: Loss Versus Rebalancing by Itô — Task

- ▣ **Question:** at what rate does a weighted pool lose against the rebalancing portfolio?
- ▣ **Model:** A5
- ▣ **Data and inputs**
 - ▶ Price $dP_t = \mu P_t dt + \sigma P_t dW_t$; pool value $V(P) = cP^w$ (A5); frictionless arbitrage, no fees [Milionis et al. \(2022\)](#)
 - ▶ The rebalancing portfolio holds the pool's Ether $V'(P_t)$ at every instant and trades at market prices:
 $dR_t = V'(P_t) dP_t$
- ▣ **Tasks**
 1. Compute V' and V'' , apply Itô's lemma to $V(P_t)$ and show $d(R_t - V_t) = \frac{1}{2}w(1-w)\sigma^2 V_t dt$.
 2. Check that $w = 1/2$ gives the lecture's rate $\sigma^2/8$.
 3. Compute the annual rate for $\sigma = 70\%$ and $w = 0.5, 0.8, 0.95$.
- ▣ **Report:** the derivation and three rates in % a year
- ▣ **Interpretation:** why does the drift μ not appear in the loss?

A8 [Proposed]: What Does a Linear AR(1) Estimate? — Task

- ▣ **Question:** what does a linear AR(1) fitted to a threshold process measure?
- ▣ **Model:** A7
- ▣ **Data and inputs**
 - ▶ The USD Coin deviations of B3; linear AR(1) $d_t = \phi d_{t-1} + e_t$ by OLS without intercept
 - ▶ $\hat{\phi}_{\text{in}}, \hat{\phi}_{\text{out}}$: the regime-wise OLS slopes at $\hat{c}$ of B3
- ▣ **Tasks**
 1. Show that, in sample, $\hat{\phi}_{\text{lin}} = (1 - s)\hat{\phi}_{\text{in}} + s\hat{\phi}_{\text{out}}$ with $s = \sum_{\text{out}} d_{t-1}^2 / \sum_t d_{t-1}^2$.
 2. Compute s for USD Coin and check the identity numerically.
 3. Compute the linear half-life and say which regime it describes.
- ▣ **Report:** s , the weighted average, $\hat{\phi}_{\text{lin}}$, the linear half-life
- ▣ **Interpretation:** why can a linear AR(1) report fast reversion although most days revert slowly?

B4 [Proposed]: Tether and Dai — Task

- ▣ **Question:** are the Tether and Dai pegs threshold processes, and which band is widest?
- ▣ **Model:** B3
- ▣ **Data and inputs**
 - ▶ Daily closes of Tether (USDT) and Dai (DAI), 1 Jan 2021 – 18 Sep 2026; deviations in bp
 - ▶ Same threshold AR, grid, bootstrap (999 draws) and level as B3
- ▣ **Tasks**
 1. Estimate $\hat{c}$, $\hat{\phi}_{\text{in}}$ and $\hat{\phi}_{\text{out}}$ with White SE for each coin.
 2. Test linearity with $\sup W$ and the bootstrap, with and without 9–16 March 2023.
 3. Plot both slopes with 95% intervals for Tether, USD Coin and Dai.
- ▣ **Report:** one table with the three coins and the chart
- ▣ **Interpretation:** which coin reverts fastest outside its band, and what would a test of two thresholds need?

B2 [Proposed]: Ether and Solana — Task

- ▣ **Question:** are Ether and Solana riskier than Bitcoin relative to shares, and do they share its weekend lull?
- ▣ **Model:** B1
- ▣ **Data and inputs**
 - ▶ Daily log returns of Ether (2018–2026) and Solana (April 2020 – 2026); S&P 500 over the same periods
 - ▶ Same split date and bootstrap settings as B1
- ▣ **Tasks**
 1. Compute each coin's annualised volatility with a block-bootstrap CI and its ratio to the S&P 500.
 2. Compute the weekend / weekday variance ratio before and after the ETFs, and a CI for the change.
- ▣ **Report:** one table with Ether, Solana and Bitcoin (B1) and a chart of the intervals
- ▣ **Interpretation:** which coin is the riskiest relative to shares?

B6 [Proposed]: ETHA and Ether — Task

- ▣ **Question:** does ETHA track Ether as well as IBIT tracks Bitcoin?
- ▣ **Model:** B5
- ▣ **Data and inputs**
 - ▶ ETHA (adjusted close) and Ether on ETHA's trading days, 23 Jul 2024 – 18 Sep 2026: 541 days, 112 weeks
 - ▶ Sponsor fee 0.25% a year; same tests and settings as B5
- ▣ **Tasks**
 1. Estimate the daily, weekly and Dimson betas.
 2. Run the ADF and KPSS tests on the log price ratio.
 3. Estimate both drifts with 95% CIs and plot the log price ratio with its trend.
- ▣ **Report:** betas with SE, test decisions, both drifts with CIs, the chart
- ▣ **Interpretation:** is ETHA a worse tracker than IBIT, or only noisier?

C1 [Proposed]: Alternative Asset, Risk Asset or Hedge? — Task

- **Question:** is Bitcoin a hedge or a safe haven for equity investors, and did the ETFs change that?
- **Model:** B1, B5
- **Data and inputs**
 - ▶ Daily log returns of Bitcoin, S&P 500, QQQ, GLD and TLT on common days, Jan 2021 – 10 Jan 2024 against 11 Jan 2024 – 18 Sep 2026
 - ▶ Hedge: uncorrelated or negatively correlated on average; safe haven: on the worst market days [Baur & Lucey \(2010\)](#); earlier evidence [Bouri et al. \(2017\)](#)
- **Tasks**
 1. Compute the correlations with Bitcoin in both periods, with block-bootstrap CIs for each and for the change.
 2. Test the change of the S&P 500 correlation with the Forbes–Rigobon adjusted correlation [Forbes & Rigobon \(2002\)](#).
 3. Estimate $r_t^{\text{BTC}} = a + b r_t^{\text{S\&P}} + \sum_q c_q r_t^{\text{S\&P}} \mathbf{1}\{r_t^{\text{S\&P}} \leq q\} + u_t$ (q : 10%, 5%, 1% quantiles, HAC SE) and quantile-regression betas at $\tau = 1\%, 5\%, 50\%$ [Koenker & Bassett \(1978\)](#).
- **Report:** a correlation table, the Forbes–Rigobon test, the regime slopes with intervals, a verdict
- **Interpretation:** is Bitcoin “alternative” in its distribution but “risk” in its co-movement?

C3 [Proposed]: Crypto Factors After July 2020 — Context and Data

- **Question:** out of sample, does three-week momentum among large coins survive, and is the crypto market still segmented from equities?
- **Model:** B5 (HAC inference), B7 (break at an unknown date); a project seed from the lecture case study
- **Context:** [Liu, Tsyvinski & Wu \(2022\)](#), building on [Liu & Tsyvinski \(2021\)](#), price 1,827 coins with CMKT, CSMB and CMOM (2014 – July 2020); three-week momentum earns 3.1% a week (Table 4)
- **Data**
 - ▶ Daily closes of Bitcoin, Ether, XRP, BNB, Cardano, Solana, Dogecoin, Litecoin, Chainlink; the lecture's five-coin market-value index as the CMKT proxy (from February 2018)
 - ▶ One-month bill rate and global five factors [Fama & French \(2015\)](#) from the [Kenneth French Data Library](#); the Canadian, Singapore, Australian dollar, euro and pound against the USD from FRED
 - ▶ Weekly simple returns on the paper's calendar (week 52 ends on 31 December); in sample to July 2020, out of sample from August 2020

C3 [Proposed]: Crypto Factors After July 2020 — Task

▣ Tasks

1. Build weekly returns for Bitcoin, XRP (from 2014) and Ether (from week 33 of 2015) to July 2020, and compare mean, median, standard deviation, skewness and kurtosis with Table 1, Panel B.
2. Regress the weekly excess return of the CMKT proxy on the five global factors and the five currencies (footnote 16 of [Liu, Tsyvinski & Wu \(2022\)](#)), in and out of sample, with HAC t -statistics and R^2 .
3. Each week, rank the coins by their past three-week return and hold the top three minus the bottom three for one week (a descriptive illustration, not the paper's CMOM); report the mean and the crypto CAPM α , β , R^2 , in and out of sample.
4. Test a change in alpha after July 2020 with a pooled HAC regression, then a break in (α, β) at an unknown date with sup-Wald, as in B7.

▣ **Report:** the comparison with the paper, the in- and out-of-sample tables, the tests of task 4, the full list of tests run

▣ **Interpretation:** if the momentum alpha of these large coins stays significant, is it a premium or the selection of coins that survived to 2026?

References (1/2)

- ▣ Adams, H., Zinsmeister, N., Robinson, D. (2020). [Uniswap v2 Core](#). Technical white paper.
- ▣ Andrews, D.W.K. (1993). [Tests for parameter instability and structural change with unknown change point](#). *Econometrica*, 61(4), 821–856.
- ▣ Athreya, K.B. (1987). [Bootstrap of the mean in the infinite variance case](#). *Annals of Statistics*, 15(2), 724–731.
- ▣ Bai, J. (1997). [Estimation of a change point in multiple regression models](#). *Review of Economics and Statistics*, 79(4), 551–563.
- ▣ Balke, N.S., Fomby, T.B. (1997). [Threshold cointegration](#). *International Economic Review*, 38(3), 627–645.
- ▣ Baur, D.G., Lucey, B.M. (2010). [Is gold a hedge or a safe haven? An analysis of stocks, bonds and gold](#). *Financial Review*, 45(2), 217–229.
- ▣ BlackRock (2026). [iShares Bitcoin Trust ETF \(IBIT\)](#). Product page.
- ▣ Bouri, E., Molnár, P., Azzi, G., Roubaud, D., Hagfors, L.I. (2017). [On the hedge and safe haven properties of Bitcoin: is it really more than a diversifier?](#). *Finance Research Letters*, 20, 192–198.
- ▣ Dickey, D.A., Fuller, W.A. (1979). [Distribution of the estimators for autoregressive time series with a unit root](#). *Journal of the American Statistical Association*, 74(366), 427–431.
- ▣ Dimson, E. (1979). [Risk measurement when shares are subject to infrequent trading](#). *Journal of Financial Economics*, 7(2), 197–226.
- ▣ Diop, P.O., Chevallier, J., Sanhaji, B. (2024). [Collapse of Silicon Valley Bank and USDC depegging: a machine learning experiment](#). *FinTech*, 3(4), 569–590.
- ▣ Efron, B. (1979). [Bootstrap methods: another look at the jackknife](#). *Annals of Statistics*, 7(1), 1–26.
- ▣ Fama, E.F., French, K.R. (2015). [A five-factor asset pricing model](#). *Journal of Financial Economics*, 116(1), 1–22.

References (2/2)

- ▣ Forbes, K.J., Rigobon, R. (2002). [No contagion, only interdependence: measuring stock market comovements](#). *Journal of Finance*, 57(5), 2223–2261.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). [Statistics of Financial Markets: An Introduction](#), 5th ed. Springer.
- ▣ Hansen, B.E. (1996). [Inference when a nuisance parameter is not identified under the null hypothesis](#). *Econometrica*, 64(2), 413–430.
- ▣ Koenker, R., Bassett, G. (1978). [Regression quantiles](#). *Econometrica*, 46(1), 33–50.
- ▣ Künsch, H.R. (1989). [The jackknife and the bootstrap for general stationary observations](#). *Annals of Statistics*, 17(3), 1217–1241.
- ▣ Kwiatkowski, D., Phillips, P.C.B., Schmidt, P., Shin, Y. (1992). [Testing the null hypothesis of stationarity against the alternative of a unit root](#). *Journal of Econometrics*, 54(1–3), 159–178.
- ▣ Liu, Y., Tsyvinski, A. (2021). [Risks and returns of cryptocurrency](#). *Review of Financial Studies*, 34(6), 2689–2727.
- ▣ Liu, Y., Tsyvinski, A., Wu, X. (2022). [Common risk factors in cryptocurrency](#). *Journal of Finance*, 77(2), 1133–1177.
- ▣ Lyons, R.K., Viswanath-Natraj, G. (2023). [What keeps stablecoins stable?](#). *Journal of International Money and Finance*, 131, 102777.
- ▣ Milionis, J., Moallemi, C.C., Roughgarden, T., Zhang, A.L. (2022). [Automated market making and loss-versus-rebalancing](#). arXiv:2208.06046.
- ▣ Newey, W.K., West, K.D. (1987). [A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix](#). *Econometrica*, 55(3), 703–708.
- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). [Are cryptos becoming alternative assets?](#). *European Journal of Finance*, 29(10), 1064–1105.
- ▣ Trimborn, S., Härdle, W.K. (2018). [CRIX an index for cryptocurrencies](#). *Journal of Empirical Finance*, 49, 107–122.